

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AUTONOMOUS DEGREE SEMESTER EXAMINATION

Subject: Design and Analysis of Algorithms	Sub Code: CS-501	Max Marks: 35
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions.
- 2. Assume suitable data if necessary.

SECTION I

ANALYTICAL & DESIGN PROBLEMS

1. Explain the fundamental working principle of Introduction: Algorithm with suitable illustrations. **[2 Marks]**
2. Discuss how Disjoint Sets: Disjoint set operations handles edge cases, boundary conditions, or exceptions. **[2 Marks]**
3. Compare and contrast the advantages and limitations of Dynamic Programming: General method. **[2 Marks]**
4. Illustrate the step-by-step mechanism of Greedy method: General method using a concrete example. **[2 Marks]**
5. Illustrate the step-by-step mechanism of Branch and Bound: General method using a concrete example. **[2 Marks]**

SECTION 2

6. (a) Describe the architectural or structural flow involved in Performance Analysis-Space complexity. **[5 Marks]**

[OR]

- (b) Design a robust solution using Introduction: Algorithm for an enterprise-scale engineering problem. Analyze its complexity. **[5 Marks]**

7. (a) Compare and contrast the advantages and limitations of union and find algorithms. **[5 Marks]**

[OR]

- (b) Explain the end-to-end design and implementation of Disjoint Sets: Disjoint set operations. Trace its execution on a non-trivial benchmark dataset. **[5 Marks]**

8. (a) Explain the fundamental working principle of applications- Optimal binary search tree with suitable illustrations. **[5 Marks]**

[OR]

- (b) Discuss the trade-offs in time, space, and scalability when engineering systems centered around Dynamic Programming: General method. **[5 Marks]**

- 9. (a)** Discuss how applications- Job sequencing with deadlines handles edge cases, boundary conditions, or exceptions. **[5 Marks]**

[OR]

- (b)** Critically evaluate Greedy method: General method against alternative paradigms. When is Greedy method: General method strictly optimal? **[5 Marks]**

- 10. (a)** Compare and contrast the advantages and limitations of applications. **[5 Marks]**

[OR]

- (b)** Design a robust solution using Branch and Bound: General method for an enterprise-scale engineering problem. Analyze its complexity. **[5 Marks]**